

Spectral Predictor

Y Labels, Eigenvalues and k_{90}

A step-by-step lesson with two fully worked examples

How to read this lesson

The Spectral Predictor takes a model's internal vectors and the true answers (Y labels). It then checks whether the model's strongest directions match the direction where the true answers actually lie.

We use one small model with **3 questions**. Everything is calculated by hand.

The lesson has two parts.

- **Steps 1 to 3** are on the model side. They do not depend on the answers.
- **Steps 4 to 6** are on the task side. They use the true answers (Y labels).

There are two examples of the true answers.

- **Example A:** answers 10, -10 , 0. The model is confused. Score = 0.25.
- **Example B:** answers 12, -6 , -6 . The answer power is spread across two directions. Score = 0.375.

Steps 1 to 3 are the same for both examples. Only Step 4 onwards changes.

The six steps of the pipeline match the Parts of this lesson like this: Step 1 = Part 1, Step 2 = Part 2, Step 3 = Part 3, Step 4 = Part 5, Step 5 = Part 6, Step 6 = Part 7. Part 4 explains U_1 and U_2 . Parts 8 to 10 explain the zeros, what the predictor watches, and the summary.

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Part 1 Model side: centering the vectors

Before the output layer, the model produced one vector for each question.

Question	Vector
Question 1 (v_1)	$[4, 2]$
Question 2 (v_2)	$[2, 4]$
Question 3 (v_3)	$[0, 0]$

The spectral pipeline first finds the average of these vectors. Then it subtracts the average from each vector. This is called **centering**.

Average of dimension 1:

$$\frac{4 + 2 + 0}{3} = 2$$

Average of dimension 2:

$$\frac{2 + 4 + 0}{3} = 2$$

The average vector is $[2, 2]$.

Subtract the average from every vector.

$$c_1 = [4 - 2, 2 - 2] = [2, 0]$$

$$c_2 = [2 - 2, 4 - 2] = [0, 2]$$

$$c_3 = [0 - 2, 0 - 2] = [-2, -2]$$

Remember

A mathematical rule: whenever we centre data (subtract the average), the data loses one dimension. That is why, for 3 questions, the Gram matrix will always have one eigenvalue equal to 0.

Part 2 Model side: the Gram matrix K

Now take the dot product of every centred vector with every other centred vector. This gives a 3×3 matrix. It is called the **Gram matrix** K .

Row 1 (c_1 with all):

$$K_{1,1} = c_1 \cdot c_1 = (2 \times 2) + (0 \times 0) = 4$$

$$K_{1,2} = c_1 \cdot c_2 = (2 \times 0) + (0 \times 2) = 0$$

$$K_{1,3} = c_1 \cdot c_3 = (2 \times -2) + (0 \times -2) = -4$$

Row 2 (c_2 with all):

$$K_{2,1} = c_2 \cdot c_1 = (0 \times 2) + (2 \times 0) = 0$$

$$K_{2,2} = c_2 \cdot c_2 = (0 \times 0) + (2 \times 2) = 4$$

$$K_{2,3} = c_2 \cdot c_3 = (0 \times -2) + (2 \times -2) = -4$$

Row 3 (c_3 with all):

$$K_{3,1} = c_3 \cdot c_1 = (-2 \times 2) + (-2 \times 0) = -4$$

$$K_{3,2} = c_3 \cdot c_2 = (-2 \times 0) + (-2 \times 2) = -4$$

$$K_{3,3} = c_3 \cdot c_3 = (-2 \times -2) + (-2 \times -2) = 4 + 4 = 8$$

$$K = \begin{bmatrix} 4 & 0 & -4 \\ 0 & 4 & -4 \\ -4 & -4 & 8 \end{bmatrix}$$

Part 3 Model side: eigen-decomposition

Eigen-decomposition of K gives two things.

- Three **strengths** (eigenvalues, λ). These tell how strongly the model is thinking.
- Three **directions** (eigenvectors, U). Each strength has one direction attached to it.

3.1 Finding the eigenvalues λ

We subtract λ from the diagonal of K . Then we set the determinant equal to zero, that is $|K - \lambda I| = 0$.

$$\det \begin{bmatrix} 4 - \lambda & 0 & -4 \\ 0 & 4 - \lambda & -4 \\ -4 & -4 & 8 - \lambda \end{bmatrix} = 0$$

Expand the determinant along the first row.

$$(4 - \lambda)[(4 - \lambda)(8 - \lambda) - (-4 \times -4)] - 0 + (-4)[(0 \times (8 - \lambda)) - (-4 \times (4 - \lambda))] = 0$$

Solve inside the brackets.

$$(4 - \lambda)[(32 - 12\lambda + \lambda^2) - 16] - 4[0 - (-16 + 4\lambda)] = 0$$

$$(4 - \lambda)(\lambda^2 - 12\lambda + 16) - 4(16 - 4\lambda) = 0$$

$$(4 - \lambda)(\lambda^2 - 12\lambda + 16) - 16(4 - \lambda) = 0$$

Take $(4 - \lambda)$ common from both parts.

$$(4 - \lambda)[\lambda^2 - 12\lambda + 16 - 16] = 0$$

$$(4 - \lambda)(\lambda^2 - 12\lambda) = 0$$

$$(4 - \lambda)\lambda(\lambda - 12) = 0$$

So the three eigenvalues are:

$$\lambda_1 = 12, \quad \lambda_2 = 4, \quad \lambda_3 = 0$$

Mode	Strength (λ)
Mode 1 (strongest)	12
Mode 2 (second)	4
Mode 3 (third)	0

3.2 Finding the eigenvectors U

For each eigenvalue, we solve $(K - \lambda I)v = 0$. Here $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

Mode 1 ($\lambda = 12$)

$$\begin{bmatrix} 4-12 & 0 & -4 \\ 0 & 4-12 & -4 \\ -4 & -4 & 8-12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} -8 & 0 & -4 \\ 0 & -8 & -4 \\ -4 & -4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Equations from the first two rows:

1. $-8x - 4z = 0 \implies 8x = -4z \implies z = -2x$
2. $-8y - 4z = 0 \implies 8y = -4z \implies z = -2y$

This means x and y must be equal. Take $x = 1$ and $y = 1$. Then $z = -2$.

Raw vector: $[1, 1, -2]$.

Length (Pythagoras): $\sqrt{1^2 + 1^2 + (-2)^2} = \sqrt{1 + 1 + 4} = \sqrt{6} \approx 2.449$.

Divide the vector by its length. This is **normalizing**.

$$U_1 = \begin{bmatrix} 1/2.449 \\ 1/2.449 \\ -2/2.449 \end{bmatrix} = \begin{bmatrix} 0.408 \\ 0.408 \\ -0.816 \end{bmatrix}$$

Mode 2 ($\lambda = 4$)

$$\begin{bmatrix} 4-4 & 0 & -4 \\ 0 & 4-4 & -4 \\ -4 & -4 & 8-4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} 0 & 0 & -4 \\ 0 & 0 & -4 \\ -4 & -4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Equations:

1. First row: $-4z = 0 \implies z = 0$.
2. Third row: $-4x - 4y + 4z = 0$. Since $z = 0$, we get $-4x - 4y = 0 \implies -4x = 4y \implies x = -y$.

Take $x = 1$. Then $y = -1$, and $z = 0$. Raw vector: $[1, -1, 0]$.

Length: $\sqrt{1^2 + (-1)^2 + 0^2} = \sqrt{2} \approx 1.414$.

$$U_2 = \begin{bmatrix} 1/1.414 \\ -1/1.414 \\ 0/1.414 \end{bmatrix} = \begin{bmatrix} 0.707 \\ -0.707 \\ 0 \end{bmatrix}$$

Mode 3 ($\lambda = 0$)

$$\begin{bmatrix} 4 & 0 & -4 \\ 0 & 4 & -4 \\ -4 & -4 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Equations:

1. First row: $4x - 4z = 0 \implies x = z$.

2. Second row: $4y - 4z = 0 \implies y = z$.

All three dimensions are equal. Take $x = 1$. Then $y = 1$ and $z = 1$. Raw vector: $[1, 1, 1]$.

Length: $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \approx 1.732$.

$$U_3 = \begin{bmatrix} 1/1.732 \\ 1/1.732 \\ 1/1.732 \end{bmatrix} = \begin{bmatrix} 0.577 \\ 0.577 \\ 0.577 \end{bmatrix}$$

3.3 Summary of the model side

Mode	Strength λ	Direction U
Mode 1	12	$[0.408, 0.408, -0.816]$
Mode 2	4	$[0.707, -0.707, 0]$
Mode 3	0	$[0.577, 0.577, 0.577]$

Part 4 What are U_1 and U_2 ?

U_1 and U_2 are the **eigenvectors** (directions of thinking). We found them in Step 3 by solving the Gram matrix K .

When we broke the matrix in Step 3, we got different strengths (eigenvalues, λ). Each strength has one direction (eigenvector, U) attached to it.

- U_1 is the direction of the strongest mode ($\lambda_1 = 12$). Its value is $[0.408, 0.408, -0.816]$.
- U_2 is the direction of the second mode ($\lambda_2 = 4$). Its value is $[0.707, -0.707, 0]$.
- U_3 is the direction of the third mode ($\lambda_3 = 0$).

What is their job in the next step?

The purpose of the next step (Task Power Matching) is only to check one thing. Along which direction (U_1 or U_2) does the true answer (Y_c) overlap perfectly?

Part 5 Task side: the true answers (Y labels)

Now the true answers enter the pipeline. There are two examples. Both start from the same model (Steps 1 to 3).

5.1 Example A: answers 10, -10, 0

Question	True answer
Question 1	10
Question 2	-10
Question 3	0

Average:

$$\frac{10 - 10 + 0}{3} = 0$$

The average is already 0. So subtracting it changes nothing. The centred answers are the same:

$$Y_c = [10, -10, 0]$$

Total power (variance) of the true answers:

$$10^2 + (-10)^2 + 0^2 = 100 + 100 + 0 = \mathbf{200}$$

5.2 Example B: answers 12, -6, -6

Here the answers are chosen so that their power spreads over both main directions.

Question	True answer
Question 1	12
Question 2	-6
Question 3	-6

Average:

$$\frac{12 - 6 - 6}{3} = \frac{0}{3} = 0$$

The average is already 0. So the centred answers are the same:

$$Y_c = [12, -6, -6]$$

Total power (variance):

$$12^2 + (-6)^2 + (-6)^2 = 144 + 36 + 36 = \mathbf{216}$$

Part 6 Task Power Matching

Now we check how much of the answers' power sits on each model direction.

For every eigenvector U_k , we do two things.

1. Take the dot product with Y_c . This gives the match score W_k .
2. Square W_k . This gives the **Power** on that direction.

The **Fraction** is the power on that direction divided by the total power of the answers. We write it as p_k .

6.1 Example A: $Y_c = [10, -10, 0]$, total power = 200

Mode 1 (U_1):

$$W_1 = (0.408 \times 10) + (0.408 \times -10) + (-0.816 \times 0) = 4.08 - 4.08 + 0 = 0$$

Power = $0^2 = 0$. Fraction $p_1 = 0/200 = \mathbf{0.0}$ (0% of task power).

Mode 2 (U_2):

$$W_2 = (0.707 \times 10) + (-0.707 \times -10) + (0 \times 0) = 7.07 + 7.07 = 14.14$$

Power = $(14.14)^2 \approx 200$. Fraction $p_2 = 200/200 = \mathbf{1.0}$ (100% of task power).

Mode 3 (U_3):

$$W_3 = (0.577 \times 10) + (0.577 \times -10) + (0.577 \times 0) = 5.77 - 5.77 + 0 = 0$$

Power = $0^2 = 0$. Fraction $p_3 = 0/200 = \mathbf{0.0}$ (0%).

Mode	W_k	Power	Fraction p_k
Mode 1	0	0	0.0
Mode 2	14.14	≈ 200	1.0
Mode 3	0	0	0.0

This step shows that the whole logic of the true answers sits on the U_2 direction.

6.2 Example B: $Y_c = [12, -6, -6]$, total power = 216**Mode 1 (U_1):**

$$W_1 = (0.408 \times 12) + (0.408 \times -6) + (-0.816 \times -6) = 4.896 - 2.448 + 4.896 = 7.344$$

This is not zero. Power = $(7.344)^2 \approx \mathbf{53.93}$. Fraction $p_1 = 53.93/216 = 0.249 \approx \mathbf{0.25}$ (25% of task power).

Mode 2 (U_2):

$$W_2 = (0.707 \times 12) + (-0.707 \times -6) + (0 \times -6) = 8.484 + 4.242 + 0 = 12.726$$

Power = $(12.726)^2 \approx \mathbf{161.95}$. Fraction $p_2 = 161.95/216 = 0.749 \approx \mathbf{0.75}$ (75% of task power).

Mode 3 (U_3):

$$W_3 = (0.577 \times 12) + (0.577 \times -6) + (0.577 \times -6) = 6.924 - 3.462 - 3.462 = 0$$

This is exactly 0 because of the centering rule. Fraction $p_3 = 0/216 = \mathbf{0.0}$ (0%).

Mode	W_k	Power	Fraction p_k
Mode 1	7.344	≈ 53.93	0.25
Mode 2	12.726	≈ 161.95	0.75
Mode 3	0	0	0.0

Here the power is spread over the two main directions.

Part 7 Alignment Score and k_{90}

7.1 Model focus

Now we convert the model's own strengths (eigenvalues) into fractions. The model's total strength is:

$$12 + 4 + 0 = 16$$

$$\text{Mode 1 focus} = 12/16 = \mathbf{0.75} \text{ (75\%)}$$

$$\text{Mode 2 focus} = 4/16 = \mathbf{0.25} \text{ (25\%)}$$

$$\text{Mode 3 focus} = 0/16 = \mathbf{0.0}$$

7.2 Alignment Score

To get the Alignment Score, multiply each mode's (Model Focus $\times$ Task Fraction). Then add them.

Example A:

$$\text{Score} = (0.75 \times 0.0) + (0.25 \times 1.0) + (0.0 \times 0.0) = 0 + 0.25 + 0 = \mathbf{0.25}$$

Example B:

$$\text{Score} = (0.75 \times 0.25) + (0.25 \times 0.75) + (0.0 \times 0.0) = 0.1875 + 0.1875 = \mathbf{0.375}$$

In Example B, the model put 75% of its focus where only 25% of the answers were. So the score fell from 1.0 (perfect) to 0.375.

7.3 What is k_{90} ?

Simple meaning of k_{90}

Start from the model's strongest directions (modes) at the top. Keep adding their task power one by one. The number of directions needed before the total task power crosses 90% (0.90) is called k_{90} .

Let us understand it step by step with Example A.

1. Arrange the modes in descending order. The spectral pipeline always lines up the directions by their eigenvalue (strength), from biggest to smallest. In Example A we had 3 modes.

- Mode 1: eigenvalue = 12 (the model's biggest focus). Task Power Fraction $p_1 = 0.0$.
- Mode 2: eigenvalue = 4 (second focus). Task Power Fraction $p_2 = 1.0$.
- Mode 3: eigenvalue = 0 (lowest focus). Task Power Fraction $p_3 = 0.0$.

2. Add the power (cumulative) and check the target. The predictor has to collect 90% (0.90) of the power. It starts checking from the top (Mode 1).

- **First direction (Mode 1).** Power received: 0.0. Total collected (cumulative): 0.0. Is the 0.90 target reached? No. So go to the next direction.
- **Second direction (Mode 2).** New power received: 1.0. Total collected: old 0.0 + new 1.0 = **1.0**. Is the 0.90 target reached? Yes, because $1.0 > 0.90$.

3. Final result. The predictor had to look at the first **2 directions** to reach its 90% target. So:

$$k_{90} = 2$$

Example B gives the same value.

- Up to Mode 1: power = 0.25. This is less than 0.90.
- Up to Mode 2: power = $0.25 + 0.75 = 1.00$. The 0.90 mark is crossed.

So here also $k_{90} = 2$.

7.4 What is the use of k_{90} in real life?

Perfect grokking. Suppose the model has learnt the exact logic. Then Mode 1 alone gets task power 1.0 (100%). The predictor does not need to go to a second direction. The target is reached at the first step. So $k_{90} = 1$.

A small k_{90} means the model has grasped the task perfectly.

Memorization (rote learning). Suppose there are 3830 questions and the model is memorizing. Then the true answer does not sit on any one direction. It is scattered over thousands of directions as 0.001, 0.005 and so on. To collect 90% of the power, the predictor has to add around 2500 modes. So $k_{90} = 2500$.

A big k_{90} means the model is confused.

Part 8 Why zeros appeared: two different reasons

The question was: is it necessary that in some cases the answer will be 0 when we do $U \text{ dot } Y_c$? (Recall that $W = U \cdot Y_c$, the dot product of a model direction and the true answer.)

The answer is both yes and no. There are two different cases.

8.1 Case 1: a zero on one direction is a guaranteed rule (Mode 3)

At the start of the pipeline we centred the data (average subtracted). By the rule of mathematics, that matrix has permanently lost one direction (dimension). Because of this, exactly one eigenvalue always came out as 0. In our example that was Mode 3, direction U_3 .

The true answer Y_c is also centred. Its total sum is 0. So when we take the dot product of a centred Y_c with the U_3 direction (the eigenvalue 0 direction), the answer is mathematically always 0.

This is a rule. It will always happen.

8.2 Case 2: zero on the other directions was only a coincidence (Mode 1 in Example A)

In our working example, the matching score on Mode 1 (W_1) was 0. All the power went to Mode 2 alone.

That was not a rule. It happened only because the true answers $Y_c = [10, -10, 0]$ were chosen deliberately. That set of numbers overlaps 100% with the U_2 direction. It was chosen so that the fractions and the calculation would be easy to understand.

Suppose the true answers were something else, like $[5, 10, -15]$. Then the dot product with both U_1 and U_2 would give some small non-zero power. Neither would be 0.

8.3 What happens in a real AI model?

Take the real model with 3830 questions. When the spectral pipeline runs, it has 3830 directions (modes).

- One direction will surely give 0, because of centering.
- On the other 3829 directions, the dot-product answer never comes out exactly 0.
- The power stays scattered in small pieces over thousands of directions (for example 0.05 on one, 0.002 on another).

During training, the real AI model does exactly this work. It collects the scattered power and starts gathering

it on a few main directions. When all the power is collected on a few directions, k_{90} falls quickly. Then we say that grokking has happened.

Part 9 What to watch in the Spectral Predictor

After the whole calculation, the Spectral Predictor mainly watches **two signs** to make its prediction. These signs appear on the graph well before the model's real test accuracy shows grokking.

9.1 Sign 1: the sudden fall of k_{90} (the collapse)

When the model is confused (memorization). In the early part of training, the power of the true answers is scattered over thousands of directions. The predictor sees $k_{90} = 3000$ being needed to collect 90% of the power. That means the model is still memorizing and remembering every question in a different way.

When real understanding comes. Just before grokking, a clean universal rule forms in the model's mind. The true answer's power stops scattering. It gathers on a few main directions.

Then k_{90} suddenly falls from 3000 to 10 or 20 (in another place the same fall is described as 3000 to 5 or 10). This sudden and sharp fall is the first sign.

9.2 Sign 2: the sharp rise of the Alignment Score (the rise)

The Alignment Score tells how much the model's focus and the true answer's power match each other.

In the beginning, it stays around 0.1 or 0.2. The model's attention is somewhere else. The answer is somewhere else.

As soon as the model leaves rote learning and grasps the rule, the score rises fast towards 1.0 (perfect match). This is the second big sign.

9.3 How the predictor is built

The Spectral Predictor does not do the 6-step calculation just once. It repeats it every short while during training, at every checkpoint. Then it draws a graph.

The final accuracy of the model on the real test (the ability to give correct answers) always rises late. But the mathematical structure inside changes much earlier. Just as the direction of the wind changes before a storm arrives, the predictor sees the sudden fall in k_{90} and the jump in alignment *well before* test accuracy reaches 90%.

By catching these two signs early, the predictor predicts: "The inside of the model has now changed. Soon it will start giving correct answers from outside also."

Part 10 Complete summary

10.1 The six steps in one place

Step	Name	What is done
1	Centering (model side)	Subtract the average vector from every question vector.
2	Gram matrix K	Dot product of every centred vector with every other. Result is a 3×3 matrix.
3	Eigen-decomposition	Find strengths (λ) from $ K - \lambda I = 0$. Find directions (U) from $(K - \lambda I)v = 0$. Normalize each direction.
4	Y labels entry	Enter the true answers. Centre them to get Y_c . Total power = $\sum Y_c^2$.
5	Task Power Matching	For each U_k , compute $W_k = U_k \cdot Y_c$. Power = W_k^2 . Fraction $p_k = \text{Power} / \text{Total power}$.
6	Alignment Score and k_{90}	Model focus = $\lambda_k / \sum \lambda$. Score = $\sum (\text{focus}_k \times p_k)$. k_{90} = number of top modes needed to collect 0.90 of the task power.

10.2 Results of both examples

	Example A	Example B
True answers Y_c	$[10, -10, 0]$	$[12, -6, -6]$
Total power	200	216
p_1, p_2, p_3	0.0, 1.0, 0.0	0.25, 0.75, 0.0
Model focus (Modes 1, 2, 3)	0.75, 0.25, 0.0	0.75, 0.25, 0.0
Alignment Score	0.25	0.375
k_{90}	2	2

10.3 Meaning of Example A's result

The model put 75% of its attention on Mode 1, where the actual task has 0% power. Where the whole logic of the task sits (Mode 2), the model gave only 25% of its attention.

By seeing the score of 0.25 (25%), the predictor concludes, without looking at test accuracy, that grokking is still far away.

10.4 Points to keep in mind

- U_1, U_2, U_3 are the eigenvectors of the Gram matrix K . They come from Step 3.
- Centering removes one dimension. So one eigenvalue is always 0, and the task power on that direction is always 0.
- A zero on any other direction is not a rule. It happened in Example A only because the answers were chosen that way.
- A small k_{90} means the model has grasped the task. A big k_{90} means the model is memorizing.
- Watch for two signs: k_{90} falling suddenly, and the Alignment Score rising towards 1.0.